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Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 26-03-2026

Real-Time Event Synchronization Engine

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

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INTRODUCTION



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- In many organizations and institutions, event management still depends heavily on manual coordination, phone calls, and repeated message sharing.
- This often leads to missed updates, confusion in scheduling, and poor participant management.
- The project is important because it provides a faster and more reliable way to manage events.
- Real-time synchronization helps all users stay updated without needing repeated manual follow-ups.
- In real-world situations, this is useful for colleges, companies, workshops, and social gatherings.
- The system allows users to create events, invite participants, and track responses easily.
- It also provides instant notifications and live updates whenever any event detail changes.
- Overall, it is a modern platform designed to improve event planning, coordination, and communication.

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PROBLEM STATEMENT



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- Existing event management methods often rely on manual communication, repeated follow-ups, and the use of multiple separate tools.

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- This makes it difficult for organizers to maintain accurate, consistent, and up-to-date event information at all times.

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- Important updates related to event schedules, locations, or participant responses may be delayed or missed completely.

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- Organizers may face issues such as delayed updates, missed invitations, and response confusion.

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- Participants may not receive event changes instantly, which can lead to miscommunication and scheduling-related issues.

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- Traditional systems also lack proper real-time collaboration and live status tracking.

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- As a result, managing events becomes time-consuming, less reliable, and harder to handle for both organizers and attendees.

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- Therefore, there is a need for a smart, reliable, and real-time event synchronization system to improve coordination and efficiency.

OBJECTIVES



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- To design and develop a real-time event synchronization system that enables users to create, schedule, update, organize, and monitor events through a centralized digital platform, thereby improving accessibility, data accuracy, usability, and overall event coordination for both organizers and participants.
- To implement features such as instant notifications, live event updates, participant invitation management, response tracking, and real-time data synchronization in order to reduce communication delays, avoid confusion, improve participant engagement, and enhance the overall efficiency and reliability of event management activities.

PROPOSED SYSTEM



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- The proposed system is a Real-Time Event Synchronization Engine that provides a centralized platform for managing all event-related activities efficiently.

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- It allows users to create, update, delete, and monitor events easily through a simple, modern, and user-friendly interface.

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- The system includes participant invitation management, RSVP tracking, and instant notifications to improve communication and coordination.

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- Real-time synchronization ensures that every event update is immediately reflected for all connected users without delay.

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- This helps solve existing issues such as missed updates, delayed communication, manual follow-ups, and participant confusion.

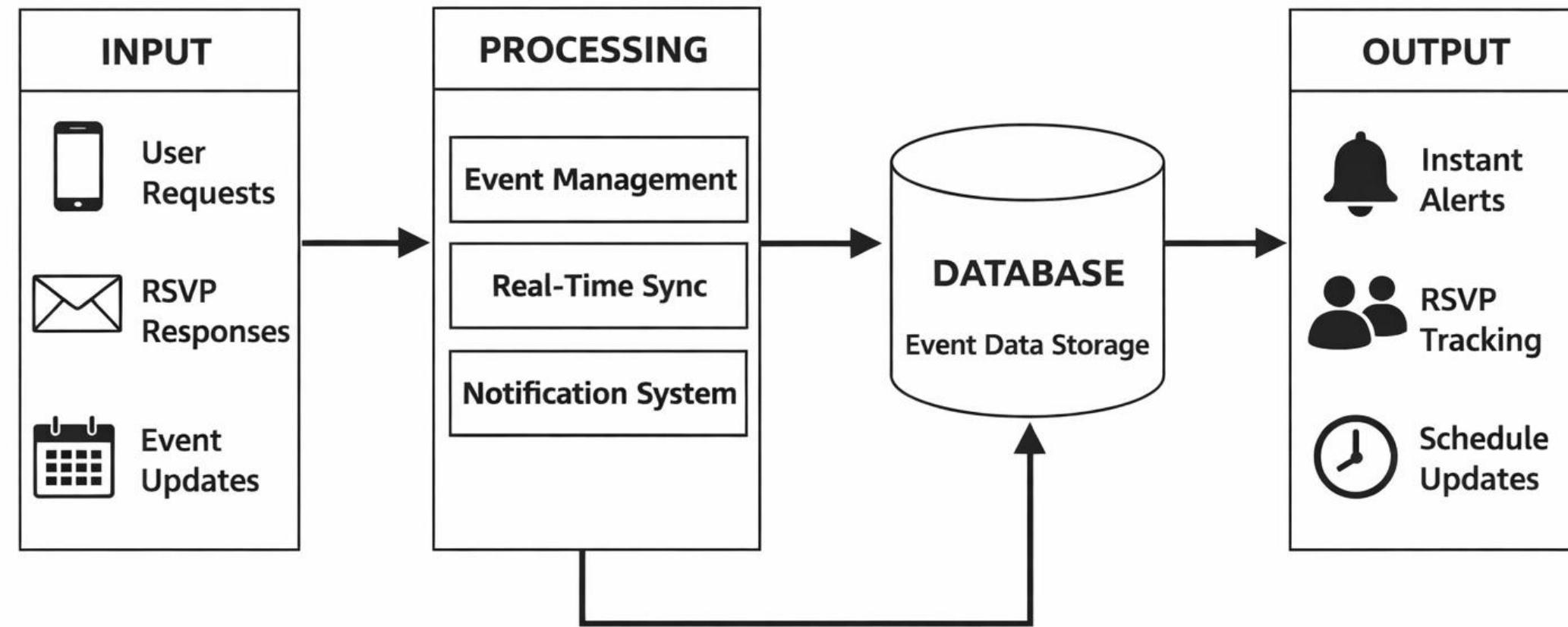
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- Compared to traditional methods, the system offers better accuracy, faster response, improved transparency, and more efficient event management.

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PROPOSED SYSTEM ARCHITECTURE

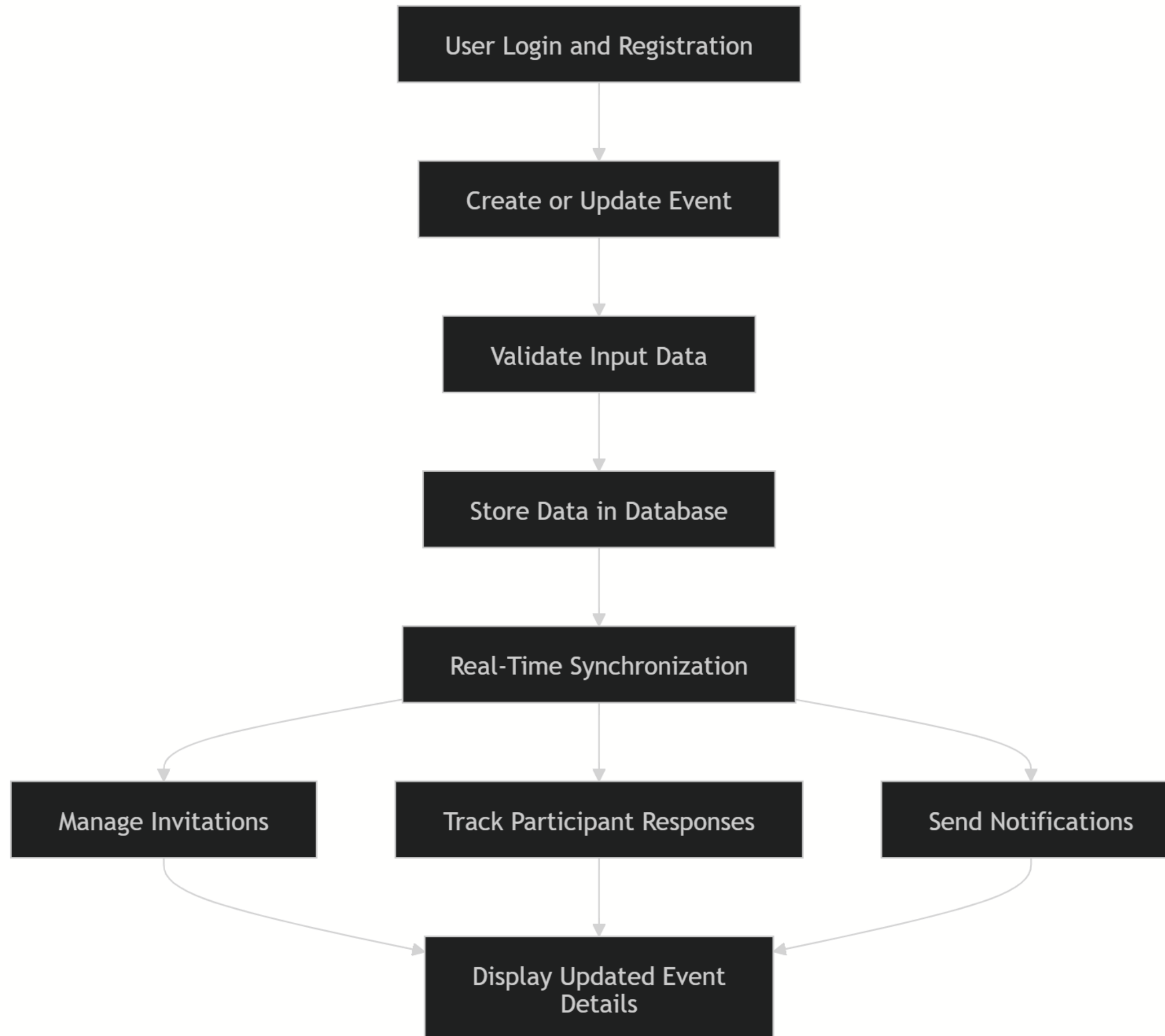


- **Input:** User data such as event details, participant invitations, responses, and event updates are entered into the system.
- **Processing:** The system processes the input by handling event creation, updating records, sending notifications, and synchronizing data in real time.
- **Database:** All information related to users, events, participants, and status is stored securely in the database for quick access and management.
- **Output:** The system displays updated event information, participant status, instant notifications, and synchronized results to all users.

WORKFLOW



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TECHNOLOGIES USED



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- **Frontend:** HTML5, CSS3, JavaScript

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- **Backend:** Node.js, Express.js, Socket.io

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- **Database:** SQLite

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- **Tools:** Visual Studio Code, Git, GitHub, Postman, npm

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- **Hardware/Software Requirements:** Computer with minimum 4 GB RAM, dual-core processor, Node.js environment, modern web browser, and operating system such as Windows, Linux, or macOS

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MODULE DESCRIPTION



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Module 1: User Authentication and Management

Handles user registration, login, secure authentication, and basic profile-related information for accessing the system.

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Module 2: Event Management

Manages event creation, updating, deletion, scheduling, and viewing of event details through the main application interface.

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Module 3: Participant and Invitation Management

Handles participant invitations, response tracking, attendee status updates, and coordination between organizers and users.

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Module 4: Real-Time Synchronization and Notification System

Ensures instant event updates, live synchronization of data, and delivery of notifications to all connected users.

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IMPLEMENTATION SCREENSHOTS



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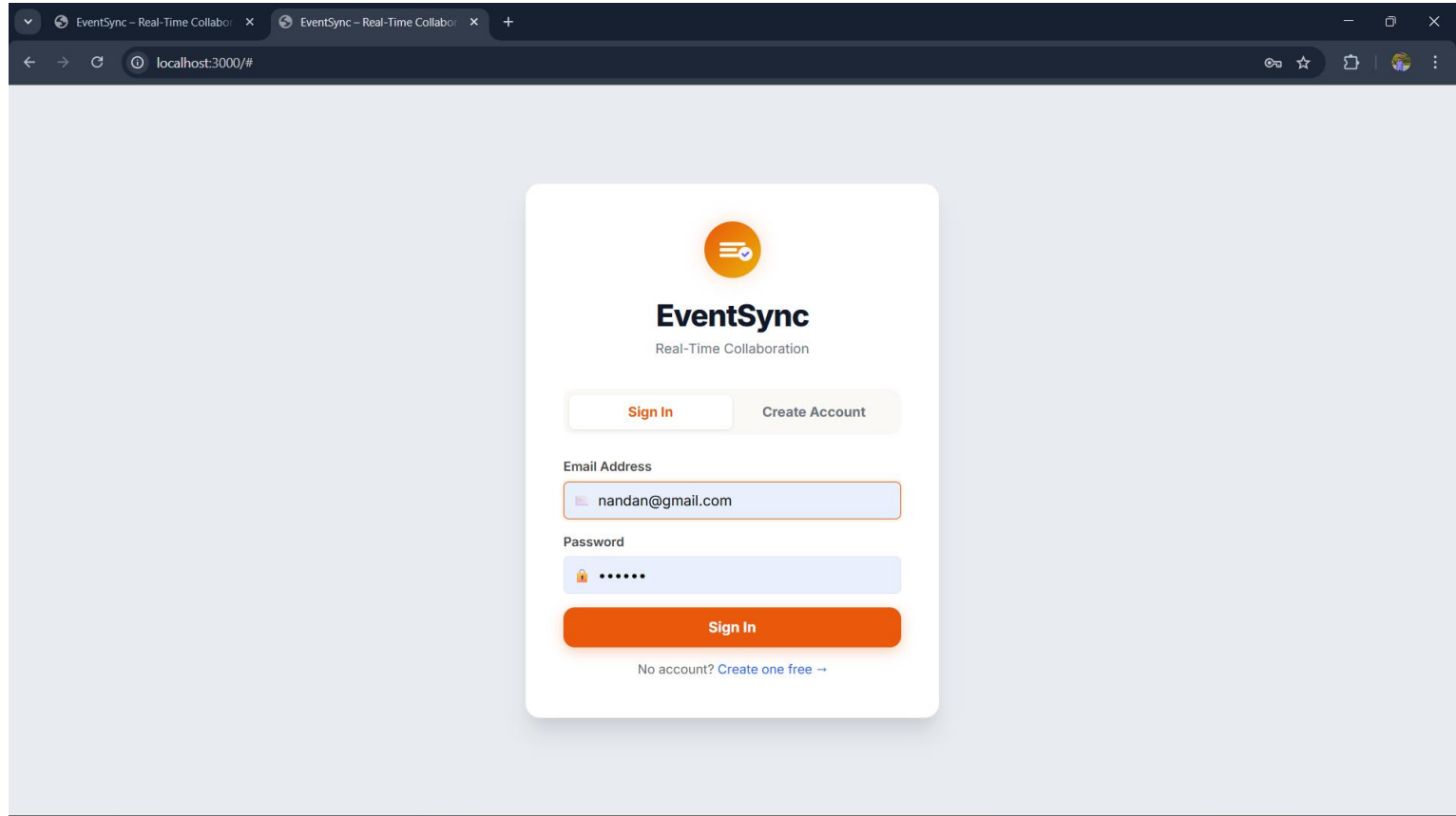


Image 1: The login page of the website

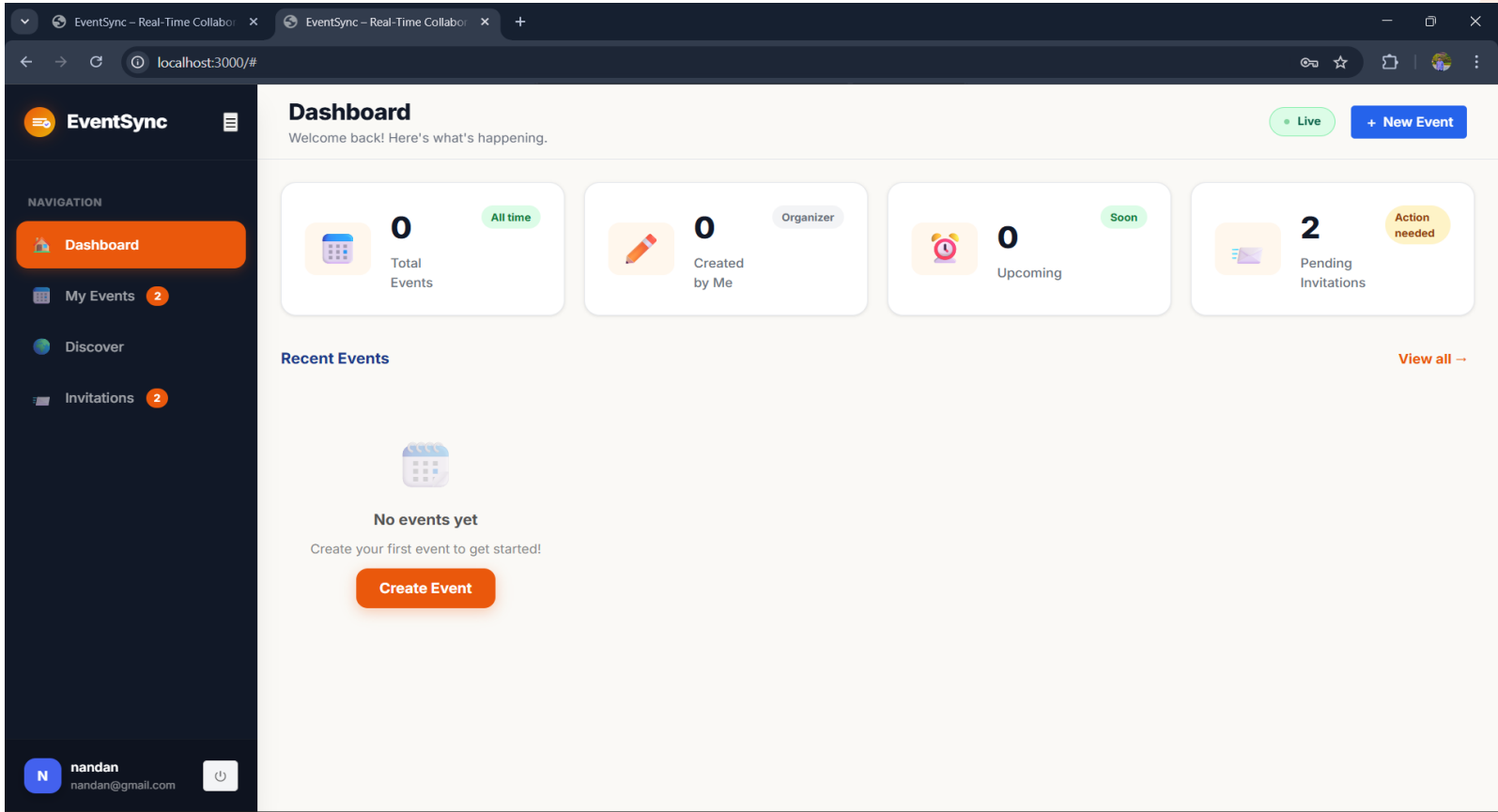


Image 2: Dashboard of the website with details regarding events

IMPLEMENTATION SCREENSHOTS



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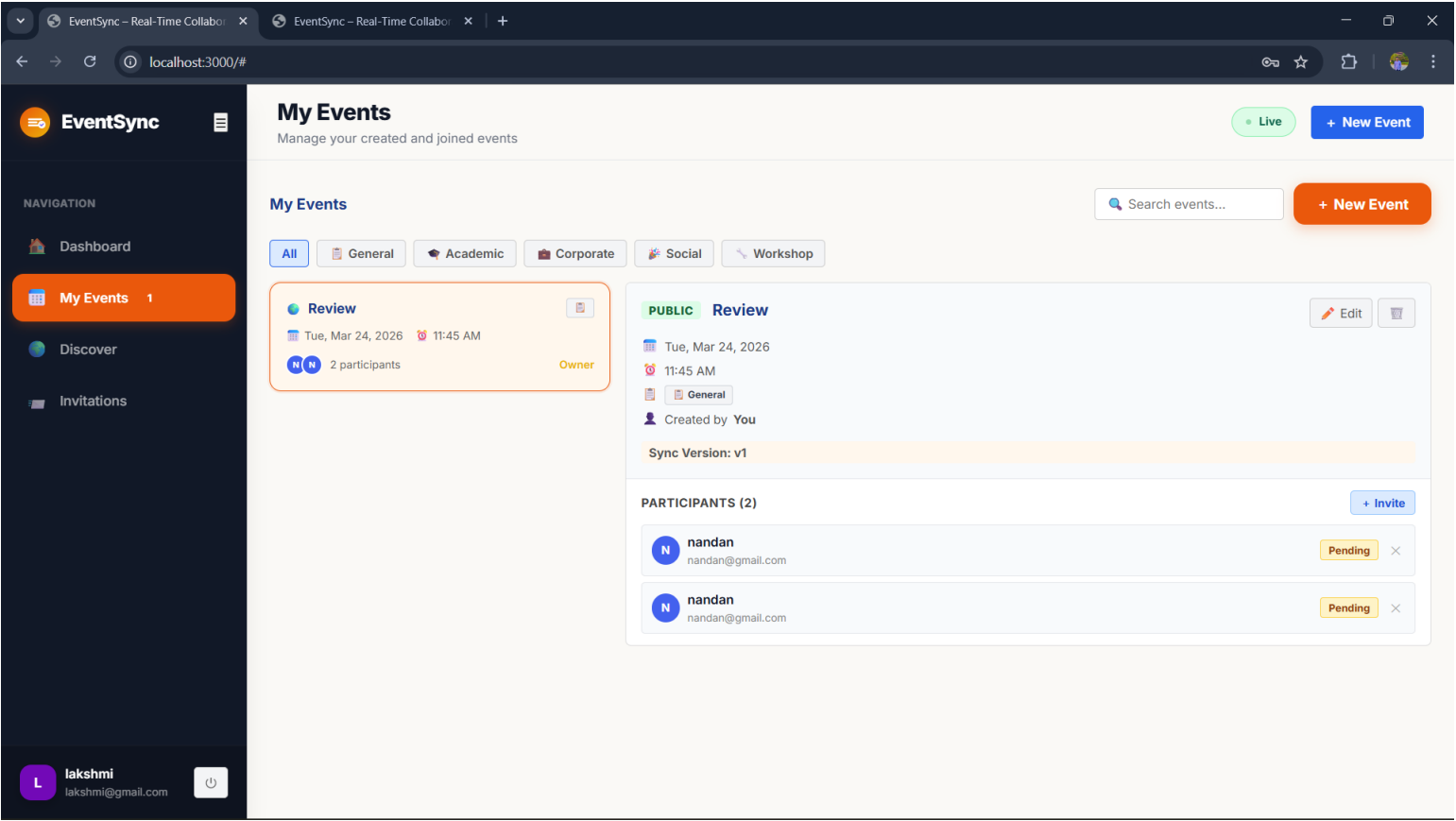


Image 3: Events screen of the user with details regarding available events.

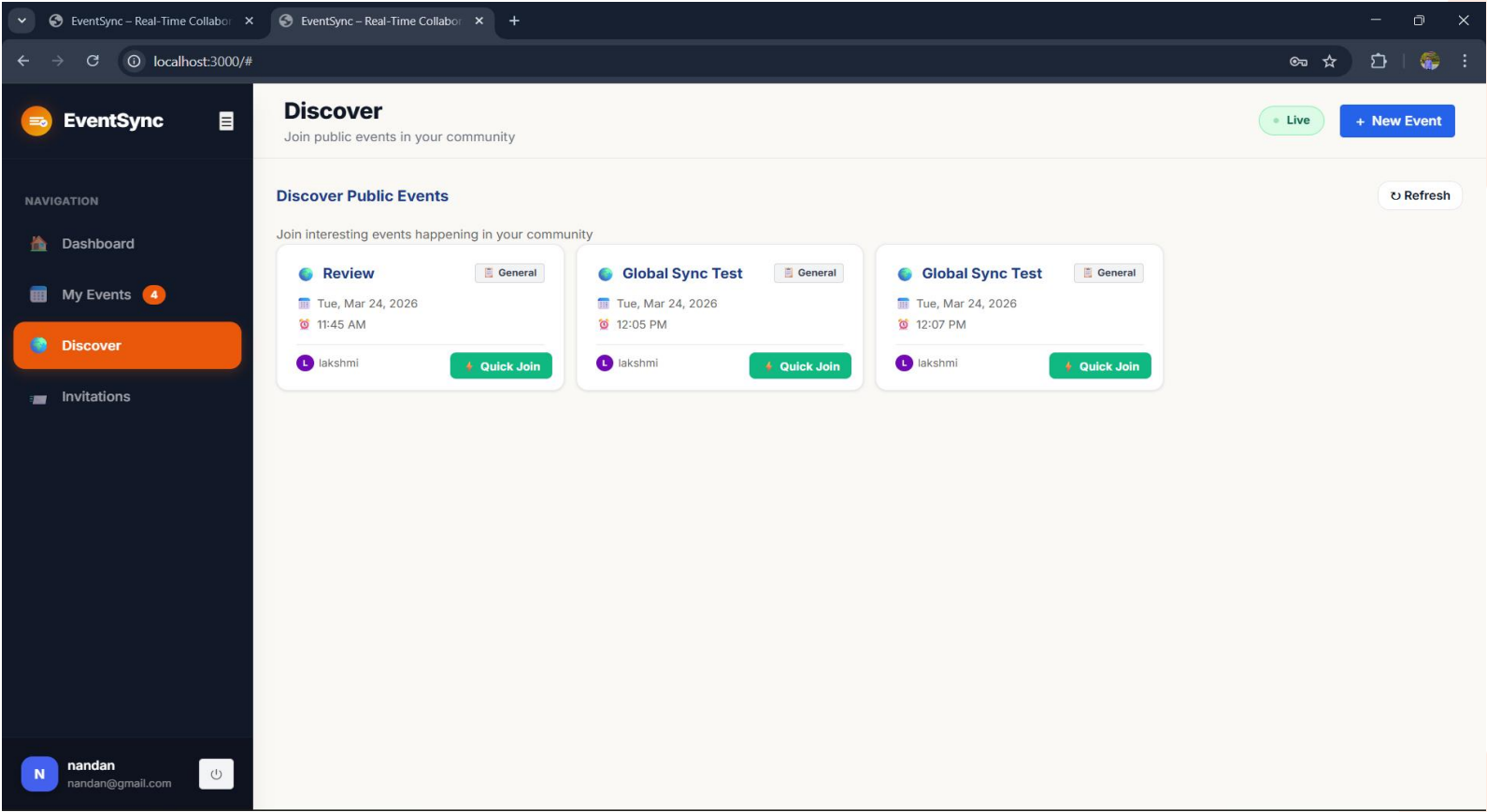


Image 4: Discover tab to explore and join various public events

IMPLEMENTATION SCREENSHOTS



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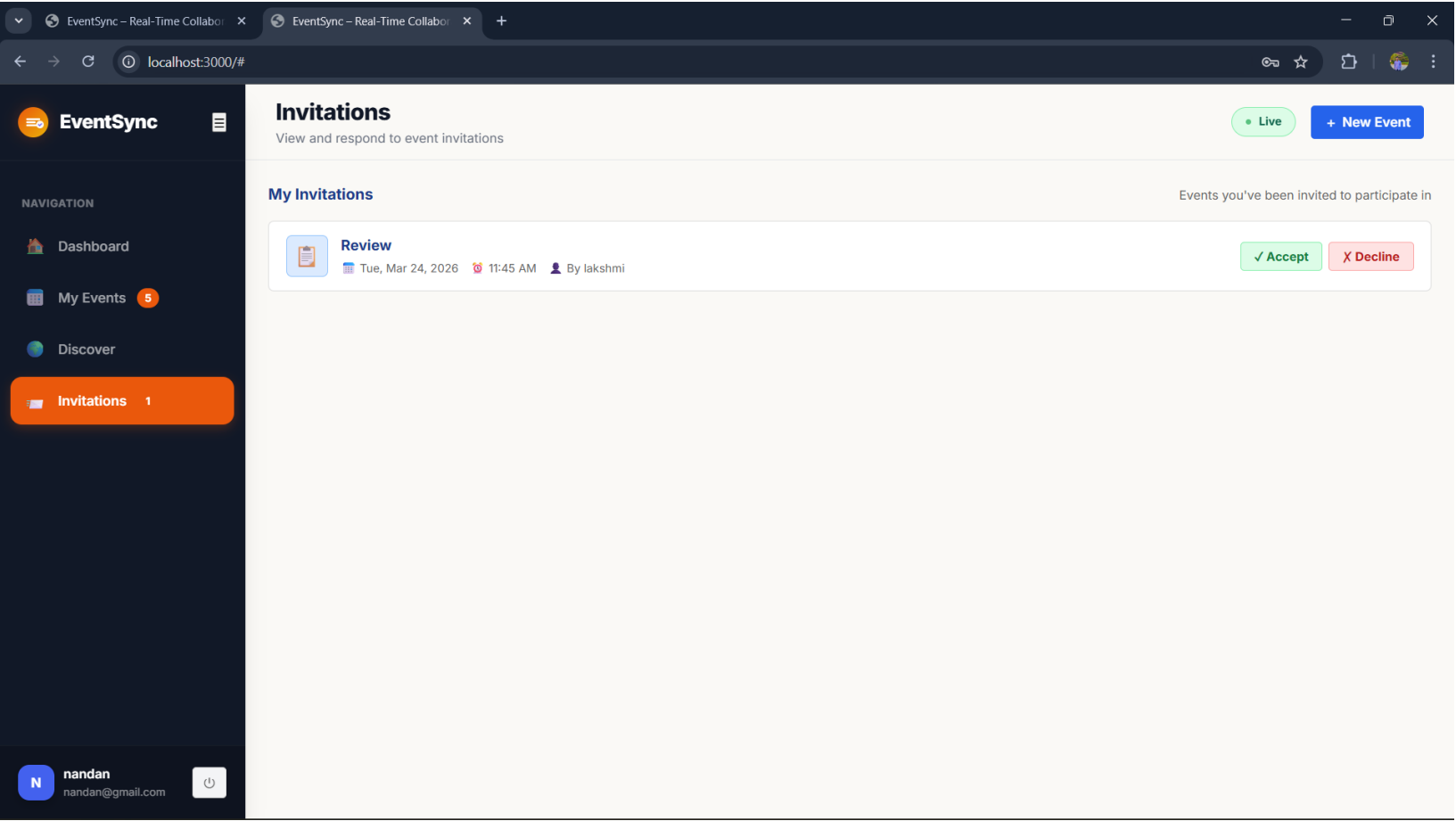


Image 5 : Invitation tab where the user can choose to join or decline an invite from another user

ADVANTAGES & LIMITATIONS



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Advantages:

- Provides real-time synchronization of event updates for all connected users.
- Reduces manual communication and improves coordination between organizers and participants.
- Supports event creation, invitation management, and live participant tracking in one platform.
- Improves accuracy, transparency, and efficiency in event handling activities.

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Limitations:

- The current system is mainly designed for small to medium-scale event management.
- Advanced analytics and reporting features are limited in the present version.
- Performance may reduce if the number of users and events increases significantly.

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Thank you

